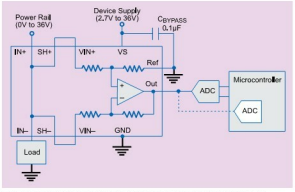

Fig. 7: Simplified schematic of INA250 from Texas Instruments. Measuring current and, thus, power is a vital housekeeping function needed in most circuits

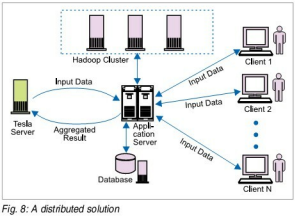

Fig. 8: A distributed solution

The combined input-referred noise voltage of the entire channel at maximum gain is 3.5nV/√Hz, which is a critical threshold parameter for effective performance.

While components and subsystems used for navigation and guidance, or for image-capture and sensing, get the most attention due to their glamour aspects, a large portion of the design of an autonomous vehicle involves mundane issues such as power management. Several application-specific, unique circuit boards and sub-systems are added to a conventional vehicle to provide the functions needed for autonomous operation. Much of the system-level operation involves measuring and managing the power requirements to control power, overall consumption and thermal dissipation.

Monitoring the current and voltage at the batteries often requires isolated sensing, for safety and functionally, but isolation is not needed on low-voltage circuit boards. Instead, the most common technique used to determine current at a source or load is with a high-side, current-sense, milli-ohm resistor (called a shunt) in conjunction with a differential amplifier that measures the voltage drop across it.

Although the amplifier is used with a discrete sense resistor, there is now an alternative that saves space, minimises errors in readings that are primarily due to the thermal drift of the sense resistor as it self-heats and simplifies the bill of materials by reducing the number of parts. INA250 from Texas Instruments puts a sense resistor and differential amplifier in a single package, resulting in a far-smaller board-layout footprint, fewer circuit-layout problems and lower system cost due to simplified schematic (Fig. 7).

The INA250 current-sense resistor plus differential amplifier components ease the design and PC-board layout tradeoffs while guaranteeing high precision and accuracy along with lower cost.

A distributed solution

The complex distributed solution included the architecture of a Web based tool for manual and semi-automated ground-truth data validation. This Web based tool included a server application, a pre-processing unit and a graphical user interface that would provide the main interface for human testers. In addition, Saguaro provided data inspection for, and delivery of,